Beginner Cloud Engineering Project (1 Week)

Project Title

Design and Deploy a Secure AWS VPC Infrastructure

Project Purpose

This project is designed to help candidates:

- Design production-ready AWS networking architecture
- Understand core cloud isolation and security practices
- Configure public and private subnets
- Implement network routing and address translation (NAT)
- Apply Network Access Control Lists (NACLs)
- Align with the AWS Well-Architected Framework principles

This bridges the gap between basic cloud theory and real-world foundational infrastructure engineering.

Duration

1 Week (5 Working Days)

Project Scenario

A startup needs a secure networking foundation hosted on AWS. The solution must support a tiered application architecture where the backend resources are completely isolated from the public internet. However, these private resources must still be able to securely download software updates and security patches. The entire setup must be securely configured, follow the principle of least privilege, and remain cost-optimized within the AWS Free Tier bounds.

Architecture Overview

Networking

- Custom VPC
- 1 Public Subnet
- 1 Private Subnet
- Internet Gateway
- NAT Gateway
- Explicit Public & Private Route Tables
- Custom Network Access Control Lists (NACLs)

Resource Naming Policy

To maintain a professional standard and ensure tracking accuracy, candidates must prepend all created resources with their initials and a project tag (e.g., `DJM-VPC-`).

1-Week Breakdown

Week 1 — Core Networking, Isolation & Security

Objectives

- Design a secure, tiered AWS VPC
- Implement secure routing logic
- Restrict network traffic using NACLs
- Document infrastructure proof

Tasks

- **VPC & Subnet Creation:** Provision a custom VPC with the CIDR block `10.0.0.0/16`. Configure both a Public Subnet (`10.0.25.0/24`) and a Private Subnet (`10.0.50.0/24`) ensuring they reside within the exact same Availability Zone (e.g., `af-south-1a`).
- **Gateways:** Deploy an Internet Gateway and attach it to your VPC. Deploy a NAT Gateway inside your Public Subnet and allocate a public Elastic IP (EIP) to it.
- **Routing Logic:** Configure a Public Route Table pointing `0.0.0.0/0` to the Internet Gateway, and associate it with the Public Subnet. Configure a Private Route Table pointing `0.0.0.0/0` to the NAT Gateway, and associate it with the Private Subnet.
- **Traffic Control:** Apply a custom Network Access Control List (NACL) to both subnets, explicitly ensuring HTTP (port 80), HTTPS (port 443), and ephemeral port ranges are permitted to allow successful outbound updates.

Deliverables

- GitHub repository with a complete `README.md` file.
- Digital architecture diagram demonstrating the layout.
- AWS Console screenshots proving correct configuration.
- Written connectivity verification theory.

Final Project Deliverables

Each candidate must submit a single GitHub repository link containing a comprehensive `README.md` file with the following items:

1. **Architecture Diagram:** A digital diagram (using Lucidchart, Draw.io, or Miro) mapping out your VPC, subnets, route tables, IGW, and NAT Gateway.
2. **Infrastructure Proof (Uncropped Screenshots):**

    - **VPC Dashboard:** Confirming the custom VPC name and the 10.0.0.0/16 CIDR block.
    - **Subnets & Route Tables:** Confirming explicit route table associations for both the public and private subnets.
    - **Gateways:** Confirming the active, available status of both the NAT Gateway and Internet Gateway.
3. **Connectivity Verification Theory:** A technical summary explaining how to verify the architecture:
    - How an EC2 instance in the Private Subnet can successfully reach the internet for updates.
    - Why an external user on the public internet cannot directly reach or access that private instance.

Learning Outcomes

By the end of this project, candidates will be able to:

- Design secure AWS networking foundations
- Establish strict network layer isolation between tiers
- Provision and map Internet and NAT Gateways properly
- Write explicit routing rules for internet inbound/outbound paths
- Configure network-level security parameters using NACLs
- Construct clean, technical documentation for stakeholders

Evaluation Rubric

Area	Weight	Description
Architecture Design	30%	Correct placement of components (NAT Gateway in Public Subnet, same AZ mapping).
Accuracy (Tech Skill)	35%	Exact matching of requested CIDR notations and accurate routing table configurations.
Security Logic	20%	Proper configuration of custom NACLs and alignment with least-privilege access rules.
Professionalism (Delivery)	15%	Adherence to naming conventions, repository cleanliness, and a well-formatted markdown presentation.

Resource Links

Here are structured learning resources for candidates:

AWS Foundations

- **AWS Free Tier:** https://aws.amazon.com/free/
- **AWS Well-Architected Framework:** https://docs.aws.amazon.com/wellarchitected/latest/framework/welcome.html
- **VPC Documentation:** https://docs.aws.amazon.com/vpc/latest/userguide/what-is-amazon-vpc.html
- **Subnets and Route Tables:** https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Subnets.html
- **NAT Gateway Documentation:** https://docs.aws.amazon.com/vpc/latest/userguide/vpc-nat-gateway.html
- **Network ACLs:** https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html

Final Outcome

At the end of this project, candidates will have:

- A production-ready AWS networking layout
- Hands-on experience navigating the AWS core console components
- A verified baseline secure network architecture
- A portfolio-ready foundation to present for technical roles

This level prepares them for:

- Cloud Infrastructure Internships
- Cloud Support Associate roles
- Basic Cloud Operator and Network Administration support
- Further practical AWS Cloud Practitioner and Solutions Architect certifications